# MOHAMMAD ABU-SHELBAIA

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## SUMMARY

Graduated with excellence from Birzeit University with a degree in Computer Engineering. Currently working as a Teaching and Research Assistant at the same institution. Experienced in full-stack web development. Proficient in Python, with a strong focus on automation, web scraping, and backend services.

## SKILLS

**Languages:** Python, LaTeX, Markdown, Bash, Assembly, Java, C, SQL, HTML, CSS

**Technologies:** Git, FastAPI, Flask, Asyncio, Django, Tailwind CSS, Docker, PostgreSQL, Arduino, Nginx, Gunicorn

## EXPERIENCE

9/2024 – Present
**Teaching And Research Assistant** — Birzeit University
Assisted in delivering laboratory sessions on microcontroller programming and embedded systems. Provided mentorship and support to students, evaluated assignments, and contributed to the preparation of lab materials.

7/2024 – 12/2024
**Backend Developer** — Silal
Contributed to the development and maintenance of a data processing pipeline for managing and analyzing product data. Wrote unit tests to ensure the reliability of Flask API endpoints.

3/2024 – 7/2024
**Backend Developer Intern** — Silal
Built a full-stack bookstore application using Django and Tailwind CSS, implementing core functionality and an intuitive user interface. Deployed the application with Docker Compose, using Nginx, Gunicorn, and PostgreSQL for a robust and production-ready setup.

## PROJECTS

Education, VitePress
Embedded, Android
**ECE Lab Manuals** — birzeit.me
Built and maintain open, step-by-step lab manuals for three Birzeit University ECE labs: ARM Cortex-M4, STM32, and Android with Spring Boot.

Web Scrapping
Python, iCal
**Course Schedule** — github.com
Developed a Python script to automate the extraction of weekly schedules from my university website and convert them into recurring iCal events. This project aimed to improve personal organization and time management by eliminating manual schedule data entry.

Django, Tailwind CSS
Nginx, Gunicorn
**Django Bookstore** — github.com
Developed a full-stack web application using Django and Tailwind CSS to display a collection of books. Utilized Nginx as a reverse proxy server and Gunicorn as the WSGI server to deploy the application using Docker''. The project aimed to showcase skills in web development and backend services.

Robotics
ESP32, Arduino
**Micromouse Robot** — github.com
Designed and implemented a maze-solving robot using the ESP32 microcontroller and the Arduino framework. The robot was programmed to autonomously navigate a maze and find the shortest path to the center using the Flood Fill algorithm.

Embedded Systems
PIC16f877A-Assembly
**PIC16f877A Calculator** — github.com
Developed a functional calculator on a PIC16f877A microcontroller, showcasing skills in embedded systems design and assembly language programming.

## EDUCATION

9/2020 - 7/2024
**Birzeit University** — Bachelor
Computer Enginering
`Cumulative Average: 87%`

28th July 2024
**Udacity** — Nanodegree Certificate
Data Analyst
`Certificate Link`

## LANGUAGES

**English** - C2 (CEFR), **Arabic** - Native